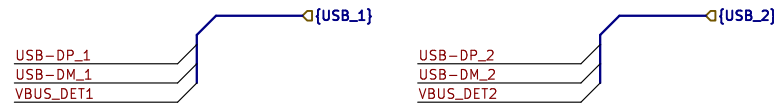
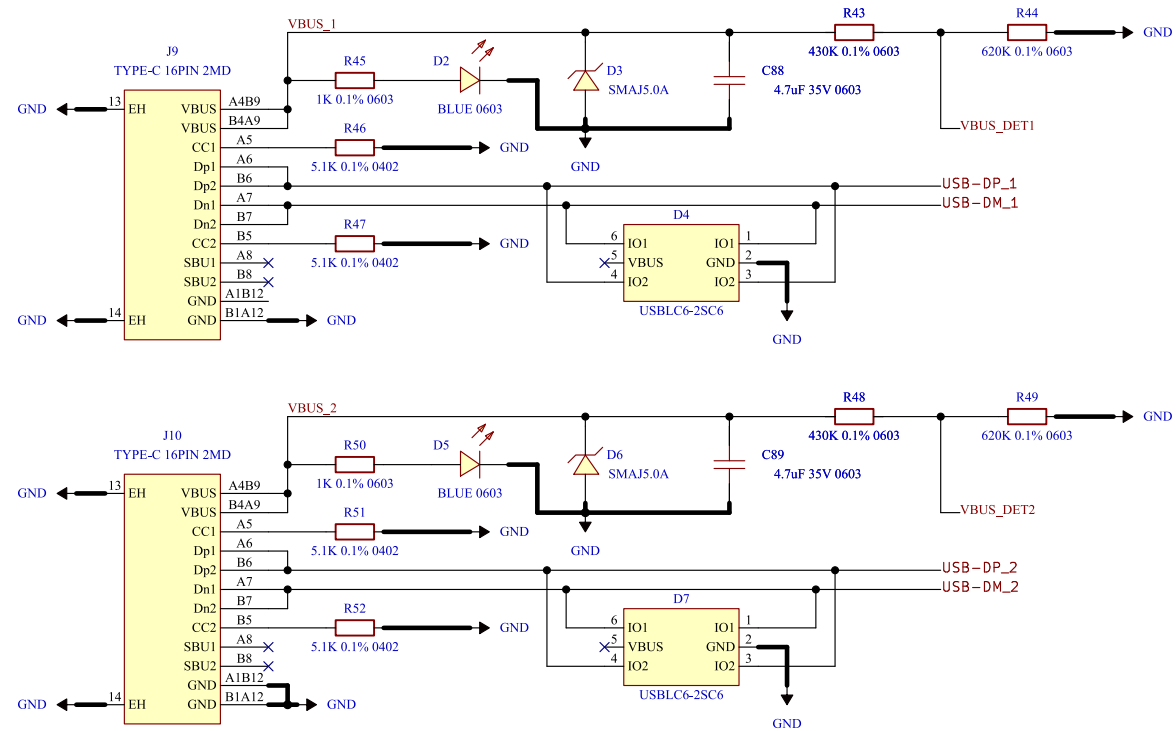
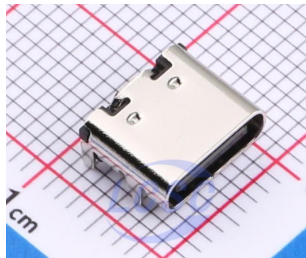
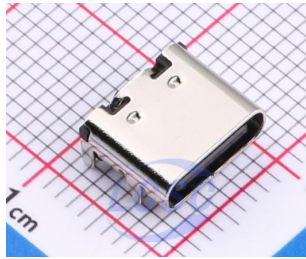


# [14] Interface - USB 2.0



## DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 §7.2.3 suspend limit2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS\_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB\_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

c99a2fe

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad\_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

Size:

A4

Sheet:

14 of 23